

Hane Lee (they/them)

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ACADEMIC POSITIONS

University of Chicago Data Science Institute

Postdoctoral Research Scientist, Data and Democracy Lab (PI: Moon Duchin)

Chicago, IL

2025 – Present

EDUCATION

Columbia University

PhD in Statistics

New York, NY

2019 – 2025

Committee: Michael Sobel, Marco Avella Medina, Naoki Egami, Robert Shapiro, Zhiliang Ying

Massachusetts Institute of Technology

MS in Media Arts and Sciences, MIT Media Lab, Opera of the Future

Cambridge, MA

2017 – 2019

BS in Electrical Engineering, minor in Music

2013 – 2017

RESEARCH INTERESTS & JOB MARKET PAPER

American politics: public opinion polarization, political preferences, electoral politics and representation.

Political methodology: measurement, latent factor models, computational methods.

Job Market Paper: [“Reconsidering the Necessity of Partisanship in Mass Polarization”](#)

Examines whether policy disagreement, specifically opinion divergence, contributes to mass polarization independently of partisan identity, using an axiomatic Wasserstein-based measure of opinion divergence.

PUBLICATIONS

Hane Lee and Michael Sobel (2026). [“Measuring Public Opinion: “The Wasserstein Bipolarization Index”, with Application to Cross-National Attitudes Toward Mandatory Vaccination for COVID-19”](#). *Annals of Applied Statistics* 20 (2) pp. 1719–1735. <https://doi.org/10.1214/26-AOAS2162>

WORKING PAPERS

Hane Lee and Andrew Davison. [“Recovering Collective Structure in Congressional Voting: A Fused Latent Factor and Network Approach”](#).

WORKING PAPERS (continued)

Yuki Atsusaka, Diana Da In Lee, and **Hane Lee**. “Quantifying Group-Based Electoral Competitiveness”.

Hane Lee and Moon Duchin. “Within-Group Coordination Under Polarized Voting: Conditions for Electoral Representation of Minority Groups.”

TEACHING EXPERIENCE

Columbia University

Instructor

Calculus-based Introduction to Statistics Summer 2024

Teaching Assistant

Graduate

Probability Theory Fall 2020, 2021, 2022, 2024; Spring 2021

Statistical Inference Fall 2022, 2023, 2024

Accelerated Probability Theory/Statistical Inference Fall 2022, 2024

Statistical Machine Learning Spring 2022

Linear Regression Models Spring 2023

Bayesian Statistics Summer 2023

Undergraduate

Introduction to Statistics / Statistical Reasoning Spring 2020 / Fall 2019

INVITED TALKS

4th Rochester Summer Conference in Applied Methods for Political Science 2026

Quantitative Methods Workshop, University of Chicago Political Science Department 2025

Structural Democracy Seminar, University of Chicago Data and Democracy Lab 2025

CONFERENCE PRESENTATIONS

2026 APSA Annual Meeting, *panel and poster presentation*

2026 Joint Statistical Meeting, *panel presentation*

2026 Society for Political Methodology Annual Conference, *co-author to panel presentation, poster*

2025 Society for Political Methodology Annual Conference, *poster presentation*

2024 Society for Political Methodology Annual Conference, *panel presentation and discussion*

2023 Society for Political Methodology Annual Conference, *poster presentation*

2022 Minghui Yu Memorial Conference, *panel presentation*

AWARDS

Howard Levine Teaching Award, Columbia University Statistics Department	2024
Columbia GSAS Travel Grant	2024
Society for Political Methodology Annual Conference Travel Grant	2023

SERVICE

PhD Student Representative, Columbia Statistics Department	2021 – 2022
Minghui Yu Memorial Conference organizer	2022
Diversity, Equity, and Inclusion Committee, Columbia Statistics Department	2020 – 2023
Department Representative, Columbia GSAS Student Council	2020 – 2021

PRIOR RESEARCH EXPERIENCE

Research Assistant, MIT Media Lab (Prof. Tod Machover)	2017-2019
Undergraduate Research Assistant, MIT RLE (Prof. George Verghese)	2016-2017
Research Intern, Ion Beam Applications (IBA) & Université Catholique de Louvain	2015
Undergraduate Research Assistant, Dr. Stefanie Shattuck-Hufnagel, MIT RLE	2014

SKILLS

Programming: *proficient* Python, R; *practical* MATLAB, javascript, C++
Languages: *native* English, Korean

REFERENCES

Moon Duchin, Professor of Computer Science and Data Science, University of Chicago
Naoki Egami, Professor of Political Science, Massachusetts Institute of Technology
Michael Sobel, Professor of Statistics, Columbia University